Journal of Machine Learning Research 27 (2026) 1-25 Submitted 10/26; Published —/—

Application of machine learning to predict the flotation of leaching tailings, Lima 2026

Edwin Miguel Barboza Duran EDWINBARBOZADURAN@GMAIL.COM
Graduate School, Master's Program in Artificial Intelligence
Universidad César Vallejo
Lima, Peru

Alfonso Ruiz Torres ALFONSO.RUIZ.TORRES@GMAIL.COM
Graduate School, Master's Program in Artificial Intelligence
Universidad César Vallejo
Lima, Peru

Editor: To be assigned

Abstract

Tree ensembles are increasingly used as soft sensors in industrial processes, but their reported accuracy often rests on random splits, implicit baselines, and predictors whose timing is not checked. We examine these issues on 547 shift-level records collected over nine months at an industrial plant that reprocesses cyanide-leached gold tailings by flotation. Random forests and XGBoost were trained on 38 process variables to estimate gold recovery, gold grade in the concentrate, and specific reagent consumption, using a chronological split, a historical-mean baseline, and the Diebold-Mariano test with the Harvey-Leybourne-Newbold correction under squared and absolute loss. On the test period, the selected learners reached R^2 values of 0.973, 0.985, and 0.897 and reduced the baseline's MAE and RMSE by more than 85%. Random forests were significantly better for recovery and XGBoost for consumption; for grade, the difference was significant only under squared loss. We show that the baseline's out-of-sample R^2 equals minus the squared standardized level shift between the fitting and test periods, which explains its strongly negative values and why a percentage-error threshold can be met by a naive predictor. We also identify two availability mechanisms—same-sample assays and additive target reconstruction—that inflate apparent accuracy.

Keywords: soft sensors, random forests, gradient boosting, temporal validation, data leakage

1 Introduction

Mine tailings are no longer viewed only as waste. Because they still contain metals that earlier processing failed to extract, reprocessing them recovers value while reducing the volume and hazard of the deposits, in line with circular-economy strategies for the mining industry (Kinnunen et al., 2022; Kursunoglu, 2025). Gold tailings that have already been leached with cyanide are a demanding case. The remaining gold is often locked in pyrite whose surfaces carry passivating films and residual cyanide, which depress flotation (Huang and Yang, 2022; Allard et al., 2022), and the feed of a plant that collects tailings from several deposits is mineralogically heterogeneous. Flotation itself is a multivariable, nonlinear process in which pH, pulp density, reagent dosage, redox potential, aeration, and particle size interact (Szmigiel et al., 2024).

©2026 Edwin Miguel Barboza Duran and Alfonso Ruiz Torres.

License: CC-BY 4.0, see https://creativecommons.org/licenses/by/4.0/. Attribution requirements are provided at http://jmlr.org/papers/v27/XX-XXXX.html.

Data-driven soft sensors—models that estimate quantities that are slow or expensive to measure from variables that are available earlier—are an attractive answer to this variability (Pural, 2023; Nasiri Abarbekouh et al., 2025). Tree ensembles are natural candidates: tabular process data of moderate size, with heterogeneous scales, thresholds, and interactions, is the regime in which they still tend to outperform deep networks (Shwartz-Ziv and Armon, 2022; Grinsztajn et al., 2022). Recent flotation studies report coefficients of determination between roughly 0.84 and 0.97 for ensemble models trained on plant or laboratory data (He et al., 2024; Ji et al., 2024; Chehreh Chelgani et al., 2024; Erkan and Günen, 2026).

Numbers of this kind are hard to interpret without knowing how they were obtained. Four aspects of the evaluation matter. First, random splits mix past and future records and hide changes in the operating regime, which yields optimistic estimates for data with temporal dependence (Cerqueira et al., 2020; Hewamalage et al., 2023). Second, a model is useful only relative to a simple reference, yet naive baselines are rarely reported (Lones, 2024). Third, two learners are usually ranked by their metrics on a single test set without a test of whether the difference exceeds sampling variability. Fourth, and most consequential for deployment, a soft sensor must use predictors that exist before the target is measured; when it does not, the evaluation suffers from leakage and overstates the accuracy that will be obtained in use (Kapoor and Narayanan, 2023). In a related spirit, Verdinelli and Wasserman (2024) show that a variable-importance parameter can be distorted by features of the joint distribution that have nothing to do with the question being asked; we argue that accuracy metrics can be distorted in a similar way by level shifts and by the timing of the predictors.

We study these four aspects on data from an industrial concentrator in the Lima region of Peru that reprocesses cyanide-leached gold tailings by flotation. The data comprise 547 shift-level records collected between January and September 2026, with 38 predictors and three targets: gold recovery, gold grade in the concentrate, and specific reagent consumption. We compare random forests (Breiman, 2001) and XGBoost (Chen and Guestrin, 2016) under a strictly chronological protocol.

Contributions. (i) We evaluate both learners with a chronological split, a refit on all pre-test data, a single evaluation on the most recent period, and an explicit historical-mean baseline. (ii) We give a closed-form expression for the out-of-sample R^2 of that baseline under a level shift (Lemma 1) and use it, together with an identity linking RMSE and R^2 (Lemma 2), to explain negative baseline values and to show that an absolute percentage-error threshold can be satisfied by a naive predictor. (iii) We compare the learners with the Diebold-Mariano test (Diebold and Mariano, 1995) with the small-sample correction of Harvey et al. (1997), under squared and absolute loss, and declare superiority only when both losses agree. (iv) We audit predictor availability and identify two mechanisms that inflate apparent accuracy—assays made on the same sample as the target, and a target that is an exact sum of predictors—and use an elementary property of random forests (Lemma 4) to explain why boosting wins on the additive target.

Related Work. Reviews of machine learning in flotation and mineral processing document a rapid growth of the field and recurring obstacles to deployment, notably scarce reliable data, few online sensors, and outliers in industrial records (Gomez-Flores et al., 2022; Estay et al., 2023; Lartey et al., 2024; Szmigiel et al., 2024; Li et al., 2025). For recovery, He et al. (2024) used 182 days of production data and a stacking ensemble, Amankwaa-Kyeremeh et al. (2024) found Gaussian-

process regression best on more than 1.3 million records from a copper rougher circuit, and Erkan and Günen (2026) found random forests best for zinc recovery. For grade, Devasahayam (2025a,b) studied interpretable and Bayesian models on small laboratory data sets, Nasiri Abarbekouh et al. (2025) proposed physics-informed networks trained on a digital twin, and several authors estimated grades from froth images (Nakhaei et al., 2022; Zhang et al., 2023; Bendaouia et al., 2024; Chen et al., 2025). Chehreh Chelgani et al. (2024) combined CatBoost with SHAP on industrial copper data and found random forests ahead of XGBoost in every test scenario. Reagent dosage and resource consumption have been modeled by Yu et al. (2025) and Pérez Cortés et al. (2025). On the methodological side, Hewamalage et al. (2023) and Cerqueira et al. (2020) discuss evaluation under temporal dependence, Lones (2024) lists common pitfalls, Kapoor and Narayanan (2023) catalogue leakage, and the literature on equal predictive accuracy has developed refinements of the Diebold-Mariano test for dependence and heteroskedasticity (Harvey et al., 2024, 2025; Coroneo and Iacone, 2025). We found no study of previously cyanided gold tailings reprocessed by flotation at industrial scale, and few flotation studies that combine a temporal protocol, an explicit baseline, and a statistical comparison of learners.

Paper Outline. Section 2 describes the setting and the two issues that motivate the study. Section 3 presents the methods. Section 4 reports the experiments. Section 5 discusses availability, the behavior of the learners, and the limits of the tests and importance measures. Section 6 concludes. Proofs are in Appendix A and a summary of the predictors is in Appendix B.

2 Setting and Two Issues

Let $X \in \mathbb{R}^{38}$ denote the vector of predictors recorded for one operating shift and Y one of the three targets. Records are indexed by time, $t = 1, \ldots, n$ with $n = 547$, in chronological order. Gold recovery is the percentage of the gold in the feed that reports to the concentrate,

$$R = \frac{C\, g_c}{F\, g_f} \times 100, \tag{1}$$

where C is the concentrate mass, g_c the gold grade of the concentrate, F the feed mass, and g_f the head grade (Concha and Bascur, 2024). The concentrate grade is $g_c = A_c/C$, with A_c the fine gold contained in the concentrate. The specific reagent consumption is the total reagent mass per tonne of feed,

$$Q = \frac{1}{F}\sum_{j=1}^{5} D_j = \sum_{j=1}^{5} d_j\,, \qquad d_j = \frac{D_j}{F}, \tag{2}$$

where D_j is the mass of reagent j (copper sulfate, sodium metabisulfite, the xanthate collectors PAX + SIPX, MIBC frother, and antifoam) and d_j its dosage in kg/t. A soft sensor is a map f learned from past records and used to estimate Y_t from X_t before Y_t is reported by the laboratory or the metallurgical balance.

The plant has three sequential stages: extraction of the tailings by repulping, cyanide detoxification, and conventional flotation. It has no grinding stage, so particle size is fixed by the incoming material (on average 89.7% passing 200 mesh in the first quarter of 2026), and liberation problems cannot be corrected in the circuit. In the first quarter of 2026, recovery averaged 79.78% with a standard deviation of 7.29 percentage points (pp) and ranged from 47.45% to 90.70%, falling

below 70% in 10% of the shifts; concentrate grade averaged 7.65 g/t (coefficient of variation 9.51%); and specific reagent consumption averaged 3.98 kg/t with a coefficient of variation of 44.84% and shift values between 1.20 and 16.88 kg/t. Decisions on dosage relied largely on operator experience and on results that arrive after the shift.

2.1 Issue 1: Temporal Regime Shift and Naive Baselines

Under a random split the mean of the test targets is close to the mean of the training targets. Under a chronological split it need not be, because the operating regime moves: the feed changes, throughput is raised, and dosage practices are revised. Consider the baseline that predicts every test record with the mean $\bar{Y}$ of the fitting records. Out-of-sample R^2 is computed with the mean of the evaluated period as reference,

$$R^2 = 1 - \frac{\sum_{t\in\mathcal{T}} (y_t - \hat{y}_t)^2}{\sum_{t\in\mathcal{T}} (y_t - \bar{y}_{\text{test}})^2}, \tag{3}$$

where $\mathcal{T}$ is the set of m test indices. The following result, proved in Appendix A, gives the baseline's value exactly.

Lemma 1 *Let* $\sigma^2_{\text{test}} = m^{-1}\sum_{t\in\mathcal{T}} (y_t - \bar{y}_{\text{test}})^2$ *and define the standardized level shift* $\delta = (\bar{y}_{\text{test}} - \bar{Y})/\sigma_{\text{test}}$. *For the constant predictor* $\hat{y}_t = \bar{Y}$, *the coefficient (3) equals* $R^2_{\text{base}} = -\delta^2$.

Three consequences follow. The baseline's R^2 is never positive; it is zero only when the two periods have the same mean. A strongly negative value is therefore not a computational error but a measurement of how far the process moved between periods. And a learner that reaches a high R^2 on the test period must track both the level shift and the variation within the period, because the reference in (3) is the test mean, which no model trained on the past can know. A second identity, also proved in Appendix A, links RMSE and R^2 for any predictor:

Lemma 2 *For any predictions* $\hat{y}_t$, $\text{RMSE} = \sigma_{\text{test}}\sqrt{1 - R^2}$, *with* R^2 *as in (3).*

Lemma 2 implies that two periods with the same R^2 can have different RMSE if their dispersion differs, which matters here because dispersion fell markedly from the training to the test period (Section 4.1). The baseline also exposes a weakness of absolute thresholds on the mean absolute percentage error (MAPE). As a toy example, let Y be normal with mean 75 and standard deviation 5. The constant predictor equal to the mean has $\text{E}|Y - 75| = 5\sqrt{2/\pi} \approx 3.99$ and hence a MAPE of roughly 5.3%, comfortably below the 10% that is often labeled excellent. For targets with large values and moderate relative variation, a MAPE threshold does not separate a useful model from a naive one.

2.2 Issue 2: Predictor Availability and Target Reconstruction

A soft sensor is useful only if its predictors are known before the target. Two mechanisms can break this requirement without leaving any trace in the metrics. The first is the use of assays made on the same physical sample, and in the same analytical cycle, as the target. A model that estimates gold grade from other assays of the concentrate is a cross-assay calibration: it can be useful, for example to replace a slow fire assay by faster determinations or to flag inconsistent results, but it cannot anticipate the grade before the concentrate exists. The second mechanism is deterministic reconstruction. By (2), the specific consumption is the sum of the five dosages; if the dosages are

predictors, a sufficiently flexible learner can reconstruct Q, and its accuracy measures its ability to represent a known function rather than to anticipate an unknown one. Kapoor and Narayanan (2023) classify both situations as leakage, since the model uses information that would not be available at the moment of a legitimate prediction. Neither can be detected from the data alone; both require auditing when each predictor becomes available. We return to them in Section 5 after presenting the results.

3 Methods

This section describes the data, the partition and baseline, the learners, the metrics and acceptance criteria, the pairwise test, and the importance measures. All computations were carried out in R 4.6.1 (R Core Team, 2026) within an interactive shiny application (Chang et al., 2026).

3.1 Data

The unit of analysis is the operating shift. Each record joins, in a single row, the readings of the SCADA/PLC system, the determinations of the metallurgical laboratory, and the reagent consumption logged for that shift; synchronizing these sources prevents the conditions of one shift from being paired with the laboratory result of another. Records from plant stoppages, start-ups, scheduled maintenance, or instrument failures were excluded, as were duplicates, values outside the physical range of the sensors, and records that could not be linked to a laboratory result. All 547 records that met these criteria were used, so the sample is a census of the valid records of the period. The data-collection form was reviewed by three domain experts and piloted on ten records before full extraction.

The 38 predictors fall into five blocks (Appendix B). (a) Feed and composition: the grades of Cu, Pb, Zn, Fe, As, and Ag, the moisture, and the percentage passing 200 mesh reported in a block of determinations that the plant labels *ccPy*; head gold grade; percent solids; insolubles; pulp density; percentage passing 200 mesh of the feed; and the particle-size parameters K30, K50, and K80. (b) Reagent dosage: copper sulfate, PAX + SIPX, MIBC, sodium metabisulfite, and antifoam, in kg/t. (c) Physicochemical: initial and final pH, redox potential, dissolved oxygen, initial and final free cyanide, and residual free cyanide. (d) Operational and hydraulic: feed mass, pulp flow, wash-water flow, pulp level, and flotation time. (e) Aeration and froth: air flow, air pressure, manifold pressure, air-valve opening, and froth height. The three targets have no missing values. Among the predictors, only residual free cyanide has missing entries (31 records, 5.7%), which were handled during preprocessing; the remaining 37 predictors are complete. The plant and its operator are not identified, for confidentiality.

3.2 Chronological Partition and Baseline

The records were split in time order into training (the 382 oldest records, 69.8%), validation (the next 82, 15.0%), and test (the 83 most recent, 15.2%). Hyperparameters were selected by the validation error of models fitted on the training block. Each selected configuration was then refitted on the 464 training and validation records, so that the most recent pre-test information was used, and evaluated once on the test block, whose records played no role in any fitting or selection step.

The baseline predicts every test record with the mean of the 464 fitting records: 78.23% for recovery, 7.575 g/t for grade, and 3.154 kg/t for consumption. It has no hyperparameters and uses

no predictor; it is equivalent to scikit-learn's DummyRegressor with the mean strategy (scikit-learn developers, 2026). We preferred it to a persistence baseline because it represents what the plant would obtain by estimating each shift with its historical average, and because it is the reference implicitly used by R^2.

3.3 Learners

A regression forest averages B trees grown on bootstrap samples, each split choosing among a random subset of predictors (Breiman, 2001):

$$\hat{f}_{\mathrm{RF}}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b\,(x).$$

XGBoost builds an additive model $\hat{y} = \sum_{k=1}^{K} f_k\,(x)$ of regression trees by stagewise minimization of a regularized objective (Friedman, 2001; Chen and Guestrin, 2016),

$$\mathcal{L} = \sum_{i} \ell\,(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega\,(f_k), \qquad \Omega(f) = \gamma T + \frac{1}{2}\lambda \parallel w \parallel^2,$$

where T is the number of leaves of a tree and w its vector of leaf values. For random forests we tuned the number of trees, the maximum tree size, the minimum leaf size, and the number of candidate predictors per split; for XGBoost, the learning rate, the maximum depth, the row subsampling rate, and the column subsampling rate per tree. We used the R packages randomForest (Breiman et al., 2024) and xgboost (Chen et al., 2026).

3.4 Metrics and Acceptance Criteria

Performance on the $m = 83$ test records was measured with R^2 as in (3), the mean absolute error $\mathrm{MAE} = m^{-1}\sum_t |y_t - \hat{y}_t|$, the root mean squared error $\mathrm{RMSE} = (m^{-1}\sum_t (y_t - \hat{y}_t)^2)^{1/2}$, and $\mathrm{MAPE} = 100\, m^{-1}\sum_t |(y_t - \hat{y}_t)/y_t|$, which is well defined here because the three targets are strictly positive. The gain over the baseline was summarized by the percentage reduction

$$\mathrm{Red}_E = 100 \times \frac{E_{\mathrm{base}} - E_{\mathrm{model}}}{E_{\mathrm{base}}}, \qquad E \in \{\mathrm{MAE}, \mathrm{RMSE}\}. \tag{4}$$

The ratio RMSE/MAE was used as a diagnostic of the error distribution (Hodson, 2022). Its range and reference values follow from an elementary lemma, proved in Appendix A.

Lemma 3 *For any errors $e_1, \ldots, e_m$, not all zero, $1 \le \mathrm{RMSE/MAE} \le \sqrt{m}$. If $e \sim N(0, s^2)$, then $\{\mathrm{E}(e^2)\}^{1/2}/\mathrm{E}|e| = \sqrt{\pi/2} \approx 1.25$; if e is Laplace with scale b, the ratio is $\sqrt{2} \approx 1.41$.*

Values well above 1.25 therefore indicate that the error is concentrated in a few records, and the upper bound is attained when a single error is nonzero.

The acceptance criteria were fixed before the analysis. A target was considered adequately predicted if at least one learner simultaneously achieved $R^2 \ge 0.75$, the lower bound of the category usually called substantial (Hair et al., 2022); $\mathrm{MAPE} \le 10\%$, the upper bound of the category usually called excellent (Zarbakhsh et al., 2024); and MAE and RMSE below those of the baseline. A reduction (4) of at least 15% in both errors was adopted as a complementary criterion of practical improvement. Because no standard fixes such thresholds for flotation, they were

checked against the performance reported in prior studies (Table 6), which ranges from 0.84 to 0.97 in R^2 for the best ensemble models. Table 1 summarizes the quantities and criteria.

Quantity	Definition	Criterion or use
Baseline	Constant prediction $\hat{y}_t = \bar{Y}$	Reference for MAE and RMSE
R^2	Equation (3), test-period mean as reference	≥ 0.75
MAE	$m^{-1}\sum_t \lvert y_t - \hat{y}_t \rvert$	Below the baseline
RMSE	$\left(m^{-1}\sum_t (y_t - \hat{y}_t)^2\right)^{1/2}$	Below the baseline
MAPE	$100\, m^{-1}\sum_t \lvert (y_t - \hat{y}_t)/y_t \rvert$	$\leq 10\%$
Red_E	Equation (4)	$\geq 15\%$ (practical improvement)
RMSE/MAE	Ratio of the two errors	Diagnostic; 1.25 if errors are normal
DM*	Equations (5)–(7), t_{m-1} reference	$p < 0.05$ under both losses

Table 1: Evaluation quantities and decision criteria. All quantities are computed on the 83 test records; $\bar{Y}$ is the mean of the 464 training and validation records.

3.5 Pairwise Comparison: The Diebold-Mariano Test

To decide whether the difference between the two learners exceeds what sampling variability would produce, we used the test of Diebold and Mariano (1995). For each test record define the loss differential

$$d_t = L\left(e_{\mathrm{RF},t}\right) - L\left(e_{\mathrm{XGB},t}\right), \qquad L(e) = e^2 \text{ or } L(e) = |e|, \tag{5}$$

where $e_{\cdot,t}$ are the prediction errors of each learner. With $\bar{d}$ the mean of the differentials and $\hat{\gamma}_0$ their variance, the statistic is

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\gamma}_0/m}}. \tag{6}$$

Because the original statistic over-rejects in small samples, Harvey et al. (1997) proposed

$$\mathrm{DM}^* = \mathrm{DM}\sqrt{\frac{m + 1 - 2h + h(h-1)/m}{m}}, \tag{7}$$

to be compared with a Student t distribution with $m - 1$ degrees of freedom; for one-step predictions, $h = 1$ and the factor reduces to $\sqrt{(m-1)/m}$. This is the version implemented in the dm.test function of the forecast package (Hyndman et al., 2026); we programmed it directly. A negative statistic indicates a smaller loss for the random forest. We declared one learner significantly superior on a target only if it had a higher R^2 and lower MAE, RMSE, and MAPE than the other, and the test gave $p < 0.05$ under both squared and absolute loss.

3.6 Variable Importance

For random forests we report the permutation importance %IncMSE, the percentage increase in out-of-bag mean squared error when a predictor is permuted (Breiman, 2001; Breiman et al., 2024); for XGBoost, the relative gain of the splits that use each predictor (Chen et al., 2026). Both are global summaries that can be biased by the scale or number of distinct values of a predictor (Strobl et al., 2007) and that split or concentrate importance among correlated predictors (Ewald et al., 2024; Verdinelli and Wasserman, 2024). We use them to check whether the predictors on which each model relies make metallurgical sense, not to draw inferential conclusions. SHAP values (Lundberg et al., 2020) were planned as a complement but are not part of the results reported here.

4 Experiments

We first describe the data and the shift between periods, then report overall and per-target performance, the pairwise comparison, and the importance measures. Unless stated otherwise, metrics refer to the test period.

4.1 Data Description and Regime Shift

Table 2 and Figure 1 summarize the targets by subset. The three periods are not equivalent, and the difference is decisive for interpreting the results. Recovery averaged 80.14% in training, 69.34% in validation, and 72.88% in test, so the most recent period operated 7.3 pp below the training period. Validation was the most irregular block, with a standard deviation of 9.03 pp and a minimum of 42.59%, whereas test had the lowest dispersion (coefficient of variation 7.0%). Concentrate grade kept a stable mean (7.58, 7.53, and 7.56 g/t) while its standard deviation fell from 0.691 to 0.356 g/t. Specific consumption showed the largest displacement: its mean fell from 3.258 to 2.672 and then to 2.015 kg/t, a drop of 38.2%, and the test median (1.895 kg/t) lies below the training first quartile (2.911 kg/t).

Target	Subset	n	Mean	SD	Min	Q1	Median	Q3	Max	CV (%)
Recovery (%)	Training	382	80.139	6.915	45.121	77.166	81.703	84.825	91.372	8.6
	Validation	82	69.340	9.028	42.593	63.438	72.262	75.990	84.012	13.0
	Test	83	72.875	5.082	56.821	70.476	74.580	76.110	80.484	7.0
Grade (g/t)	Training	382	7.584	0.691	5.580	7.171	7.550	7.997	9.639	9.1
	Validation	82	7.531	0.508	6.010	7.300	7.571	7.908	8.292	6.7
	Test	83	7.561	0.356	6.696	7.313	7.477	7.770	8.861	4.7
Consumption (kg/t)	Training	382	3.258	0.632	1.157	2.911	3.239	3.586	7.027	19.4
	Validation	82	2.672	0.793	0.662	2.143	2.727	3.245	4.235	29.7
	Test	83	2.015	0.550	0.732	1.630	1.895	2.519	3.013	27.3

Table 2: Descriptive statistics of the targets by chronological subset. SD: standard deviation; Q1 and Q3: first and third quartiles; CV: coefficient of variation.

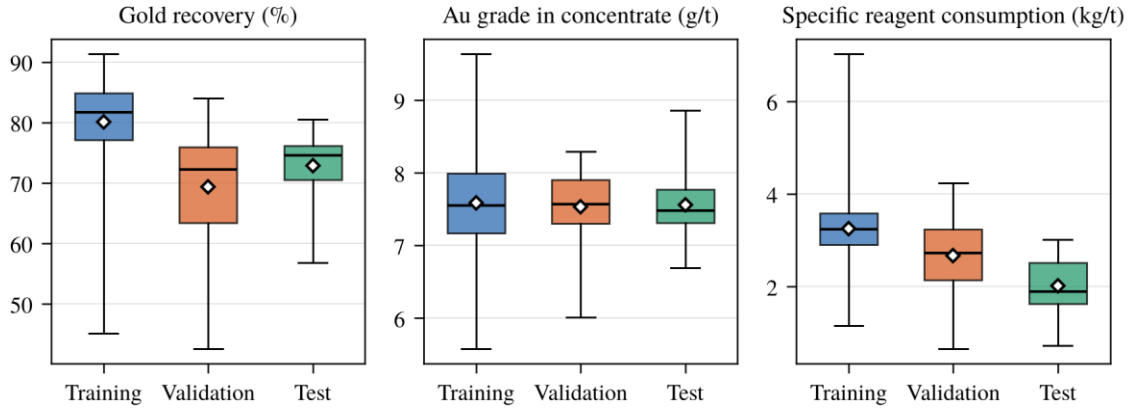


Figure 1: Distribution of the three targets by chronological subset. Boxes span the first to third quartile, the line marks the median, the diamond the mean, and the whiskers the minimum and maximum (Table 2).

The predictors moved as well (Appendix B, Table 7). Average feed mass rose from 472.0 t in training to 613.2 t in validation and 695.2 t in test, and pulp flow from 46.5 to 66.4 m³/h. Flotation time fell from 149.7 to 75.9 min and its standard deviation from 81.0 to 8.4 min. Sodium metabisulfite dosage fell from 0.974 to 0.347 kg/t, PAX + SIPX from 0.360 to 0.211 kg/t, and copper sulfate from 1.819 to 1.368 kg/t, while redox potential moved from −3.4 to 31.0 mV and final free cyanide from 30.3 to 14.3 ppm. Head grade, the particle-size parameters, and the aeration and froth variables stayed essentially constant. The test period therefore corresponds to a regime of higher throughput, shorter residence time, and lower dosage than the one that dominates the fitting data, and evaluating on it requires the models to generalize to new operating conditions. Dispersion fell from training to test in 36 of the 38 predictors. Several predictors varied by less than 5% of their mean in training (K30, K50, the ccPy percentage passing 200 mesh, pulp level, air flow, pulp density, and pH); a variable that barely changes carries little information for a model even when its physical effect is real, and none of these appears among the five most important predictors of any target.

4.2 Test-Period Performance

Table 3 reports the test metrics of the three models. With the best learner for each target, R^2 was 0.973 for recovery (random forest), 0.985 for grade (random forest), and 0.897 for consumption (XGBoost), with MAPE of 0.94%, 0.46%, and 8.92%. In the three cases MAE and RMSE fell by between 86.1% and 88.6% relative to the baseline, far above the complementary 15% criterion. Five of the six learner–target pairs met the four criteria simultaneously; the exception was the random forest on consumption, whose R^2 of 0.740 is below 0.75 and whose MAPE of 14.70% exceeds 10%. All three targets were therefore adequately predicted under the pre-specified rule.

Target	Model	R^2	MAE	RMSE	MAPE (%)	Red. MAE	Red. RMSE	Meets
Recovery (%)	Baseline	−1.124	5.473	7.362	8.07	—	—	—
	Random forest	**0.973**	**0.686**	**0.836**	**0.94**	**87.5**	**88.6**	**Yes**
	XGBoost	0.960	0.838	1.005	1.15	84.7	86.4	Yes
Grade (g/t)	Baseline	−0.002	0.285	0.354	3.77	—	—	—
	Random forest	**0.985**	**0.035**	**0.044**	**0.46**	**87.9**	**87.6**	**Yes**
	XGBoost	0.979	0.040	0.051	0.53	86.0	85.5	Yes
Consumption (kg/t)	Baseline	−4.350	1.140	1.264	70.95	—	—	—
	Random forest	0.740	0.226	0.279	14.70	80.1	78.0	No
	XGBoost	**0.897**	**0.142**	**0.175**	**8.92**	**87.6**	**86.1**	**Yes**

Table 3: Test-period performance (m = 83). Bold marks the selected learner for each target. Red. MAE and Red. RMSE: percentage reduction relative to the baseline, Equation (4). Meets: all four criteria of Table 1 satisfied. MAE and RMSE are in pp, g/t, or kg/t.

The baseline obtained R^2 values of −1.124 for recovery, −0.002 for grade, and −4.350 for consumption. Lemma 1 accounts for them exactly. For recovery, the fitting mean (78.23%) exceeds the test mean (72.88%) by 5.36 pp, and the test standard deviation in its population form is 5.05 pp, so $\delta = -1.06$ and $-\delta^2 = -1.124$. For consumption the fitting mean exceeds the test mean by 1.139 kg/t, more than two test standard deviations ($\delta = -2.08$), which reproduces the value up to rounding of the summary statistics; for grade the two means nearly coincide and $\delta \approx -0.04$ (Figure 2). The baseline also satisfied the MAPE threshold by itself for recovery (8.07%) and grade (3.77%), as anticipated in Section 2.1; for these two targets the MAPE criterion did not discriminate, and the separation came from R^2 and from error reductions of at least 84.7% for both learners. For consumption, MAPE did discriminate: 70.95% for the baseline, 14.70% for the random forest, and 8.92% for XGBoost.

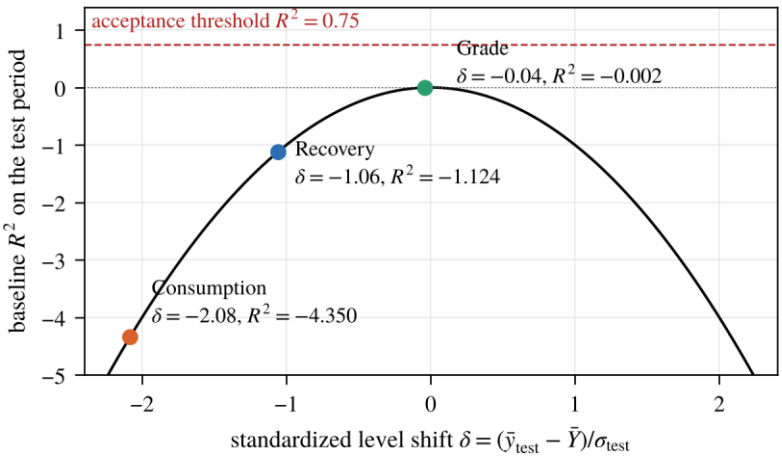


Figure 2: Out-of-sample R^2 of the historical-mean baseline as a function of the standardized level shift δ (Lemma 1). Points mark the three targets, with δ computed from the rounded summary statistics of Table 2 and the fitting means of Section 3.2; labels show the R^2 values of Table 3. The dashed line is the acceptance threshold for the learners.

Lemma 2 provides a consistency check on the reported metrics. For the random forest on recovery, $\sigma_{\text{test}}\sqrt{1 - R^2} = 5.05 \times \sqrt{0.027} = 0.830$, against a reported RMSE of 0.836; for the random forest on grade, $0.354 \times \sqrt{0.015} = 0.043$ against 0.044; and for XGBoost on consumption, $0.547 \times \sqrt{0.103} = 0.175$ against 0.175. The small discrepancies come from rounding R^2 to three decimals. The ratio RMSE/MAE lay between 1.20 and 1.29 for the six learner–target pairs (1.22 and 1.20 for recovery, 1.27 and 1.29 for grade, 1.23 and 1.24 for consumption, for the random forest and XGBoost respectively), close to the value 1.25 expected under normal errors, so the total error was not concentrated in a few large misses. As a non-decisive check against operating tolerances defined by the authors (MAE ≤ 3 pp and RMSE ≤ 5 pp for recovery; MAE ≤ 10% and RMSE ≤ 15% of the observed mean for grade and consumption), five of the six pairs complied; the exception was again the random forest on consumption, whose MAE equals 11.24% of the mean.

4.3 Results by Target

Table 4 reports the metrics of both learners in the three subsets. Training and validation metrics correspond to models fitted on the training block; test metrics to models refitted on training and validation. Figure 3 shows observed against predicted test values for the selected learners.

Target	Subset	n	Model	R^2	MAE	RMSE	MAPE (%)
Recovery (%)	Training	382	Random forest	0.995	0.348	0.491	0.45
			XGBoost	0.984	0.701	0.880	0.86
	Validation	82	Random forest	0.947	1.161	2.063	1.98
			XGBoost	0.968	1.095	1.608	1.75
	Test	83	**Random forest**	**0.973**	**0.686**	**0.836**	**0.94**
			XGBoost	0.960	0.838	1.005	1.15
Grade (g/t)	Training	382	Random forest	0.999	0.014	0.019	0.19
			XGBoost	1.000	0.001	0.002	0.02
	Validation	82	Random forest	0.992	0.035	0.045	0.47
			XGBoost	0.988	0.045	0.055	0.61
	Test	83	**Random forest**	**0.985**	**0.035**	**0.044**	**0.46**
			XGBoost	0.979	0.040	0.051	0.53
Consumption (kg/t)	Training	382	Random forest	0.957	0.071	0.130	2.29
			XGBoost	0.983	0.059	0.083	1.84
	Validation	82	Random forest	0.710	0.310	0.425	17.42
			XGBoost	0.816	0.237	0.338	13.60
	Test	83	Random forest	0.740	0.226	0.279	14.70
			XGBoost	**0.897**	**0.142**	**0.175**	**8.92**

Table 4: Performance by subset. Training and validation: models fitted on the training block; test: models refitted on training and validation. Bold marks the selected learner.

Gold recovery. The random forest reached $R^2 = 0.973$, MAE = 0.686 pp, and RMSE = 0.836 pp, about one ninth of the baseline RMSE (7.362 pp); XGBoost reached 0.960, 0.838 pp, and 1.005 pp. Training R^2 was higher (0.995 and 0.984) but the drop to test was moderate, about 0.02 for both learners, with no sign of overfitting that would compromise generalization. On validation, however, XGBoost was better (0.968 against 0.947, and RMSE 1.608 against 2.063 pp), and the order reversed on test. The validation block contained the lowest recoveries of the series, and there the random forest's RMSE/MAE ratio rose to 1.78, a sign that a few atypical shifts concentrated much of its error. In Figure 3a the predictions follow the diagonal over the whole test range, from 56.8% to 80.5%; the largest deviations, of about 2 pp, occur in isolated records between 62% and 70%, without systematic over- or underestimation at the extremes.

Concentrate grade. The random forest reached $R^2 = 0.985$, MAE = 0.035 g/t, and RMSE = 0.044 g/t, one eighth of the baseline RMSE (0.354 g/t); XGBoost reached 0.979, 0.040, and 0.051 g/t. The training fit was almost perfect, especially for XGBoost ($R^2 = 1.000$, MAE = 0.001 g/t), which essentially memorized the training records; the drop to 0.979 on test was nevertheless small. Unlike recovery, the random forest was better on both validation and test, so the ordering was stable. In Figure 3b the points cluster on the diagonal between 6.7 and 8.3 g/t, and the only test record above 8.5 g/t (8.86 g/t) was estimated within 0.1 g/t.

Specific reagent consumption. XGBoost reached $R^2 = 0.897$, MAE = 0.142 kg/t, RMSE = 0.175 kg/t, and MAPE = 8.92%, meeting all criteria; the random forest reached 0.740 and 14.70% and failed two of them, although it still reduced the baseline errors by 80.1% (MAE) and 78.0% (RMSE). This target showed the largest gap between training and test: the random forest went from 0.957 to 0.710 and 0.740, and XGBoost from 0.983 to 0.816 and 0.897, with test RMSE more than twice the training RMSE for both. The loss of accuracy coincides with the downward shift of Table 2: the test period concentrated consumptions that were rare in the fitting data. Figure 3c shows that XGBoost overestimated consumptions below 1.5 kg/t—the two lowest values, 0.73 and 0.80 kg/t, were predicted as 1.11 and 1.33 kg/t—whereas above 2.5 kg/t several predictions fell slightly below the observed values.

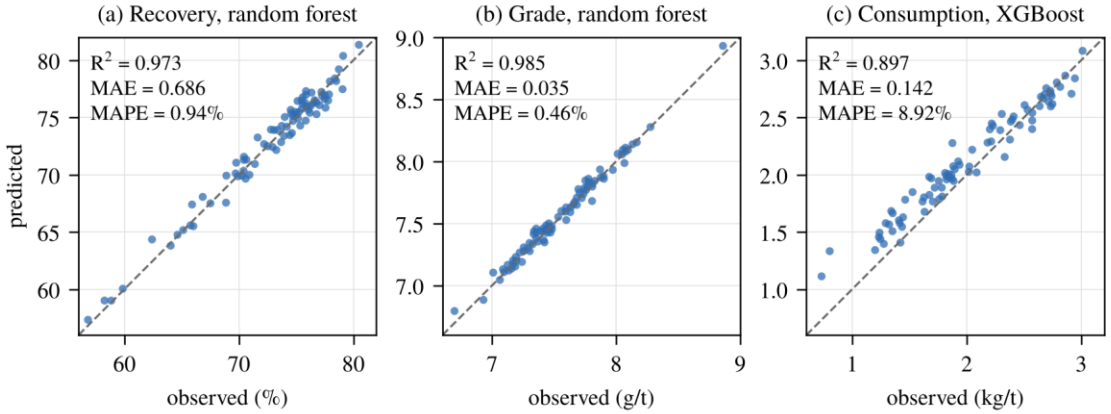


Figure 3: Observed against predicted values on the test period (m = 83) for the selected learner of each target. The dashed line is the identity; annotated metrics are those of Table 3.

4.4 Pairwise Comparison

Table 5 reports the Diebold-Mariano comparison, computed from the 83 loss differentials (5) with both loss functions and the correction (7), with 82 degrees of freedom. For recovery, the random forest dominated the four metrics and the test was significant under both losses ($\mathrm{DM}^* = -2.778$, $p = 0.0068$, with squared errors; $\mathrm{DM}^* = -2.504$, $p = 0.0143$, with absolute errors), so the random forest was significantly superior. For consumption, XGBoost dominated the four metrics and the difference was significant under both losses ($\mathrm{DM}^* = 6.070$ and 8.664, $p < 0.001$), so XGBoost was significantly superior. For grade, the random forest also dominated the four metrics and the test was significant with squared errors ($\mathrm{DM}^* = -2.174$, $p = 0.0326$) but not with absolute errors ($\mathrm{DM}^* = -1.774$, $p = 0.0797$); under the double-loss rule no significant superiority was declared.

Target	Model	R^2	MAE	RMSE	MAPE (%)	DM* (e^2)	DM* ($\lvert e \rvert$)
Recovery (%)	Random forest	0.973	0.686	0.836	0.94	−2.778 (0.0068)	−2.504 (0.0143)
	XGBoost	0.960	0.838	1.005	1.15		
	Decision					**Random forest superior**	
Grade (g/t)	Random forest	0.985	0.035	0.044	0.46	−2.174 (0.0326)	−1.774 (0.0797)
	XGBoost	0.979	0.040	0.051	0.53		
	Decision					**No significant superiority**	
Consumption (kg/t)	Random forest	0.740	0.226	0.279	14.70	6.070 (<0.001)	8.664 (<0.001)
	XGBoost	0.897	0.142	0.175	8.92		
	Decision					**XGBoost superior**	

Table 5: Diebold-Mariano comparison of random forest and XGBoost on the test period (m = 83). DM*: statistic with the Harvey-Leybourne-Newbold correction; two-sided p values in parentheses. A negative statistic indicates a smaller loss for the random forest.

No learner was best for every target. The selected configuration combines the random forest for recovery and grade with XGBoost for consumption; XGBoost was the only learner that met the criteria for all three targets.

4.5 Variable Importance

Figure 4 shows the ten most important predictors of each model. For recovery, both learners agree on the first five predictors and on their order: copper sulfate, percent solids, and the Zn, Pb, and Cu grades of the ccPy block. In XGBoost these five account for 99.7% of the total gain. In the random forest importance is more spread: copper sulfate leads with a %IncMSE of 19.8, and dissolved oxygen, final free cyanide, insolubles, the feed fineness, and the specific consumption follow with values between 1.9 and 2.3. The prominence of copper sulfate, the sulfide activator, is consistent with the role of copper activation in the flotability of auriferous pyrite and with its interplay with metabisulfite depression (Castellón et al., 2022, 2024).

For grade the pattern is different. The Fe grade of the ccPy block dominates both learners (%IncMSE of 68.9; 56.8% of the gain), followed by the As grade and moisture of the same block, the head gold grade, and K80; these five account for 99.3% of the XGBoost gain. Dosage, physicochemical, and aeration variables play a minimal role. For consumption, sodium metabisulfite and copper sulfate rank first in both learners (%IncMSE of 65.7 and 24.6; 36.4% and 25.0% of the gain), followed by PAX + SIPX, MIBC, antifoam, the Cu, Pb, and Zn grades of the ccPy block, flotation time, and percent solids; the five dosages account for 73.9% of the XGBoost gain. The mean consumption equals the sum of the five mean dosages in every subset (3.258, 2.672, and 2.014 kg/t, against an observed 2.015 kg/t on test), as (2) requires.

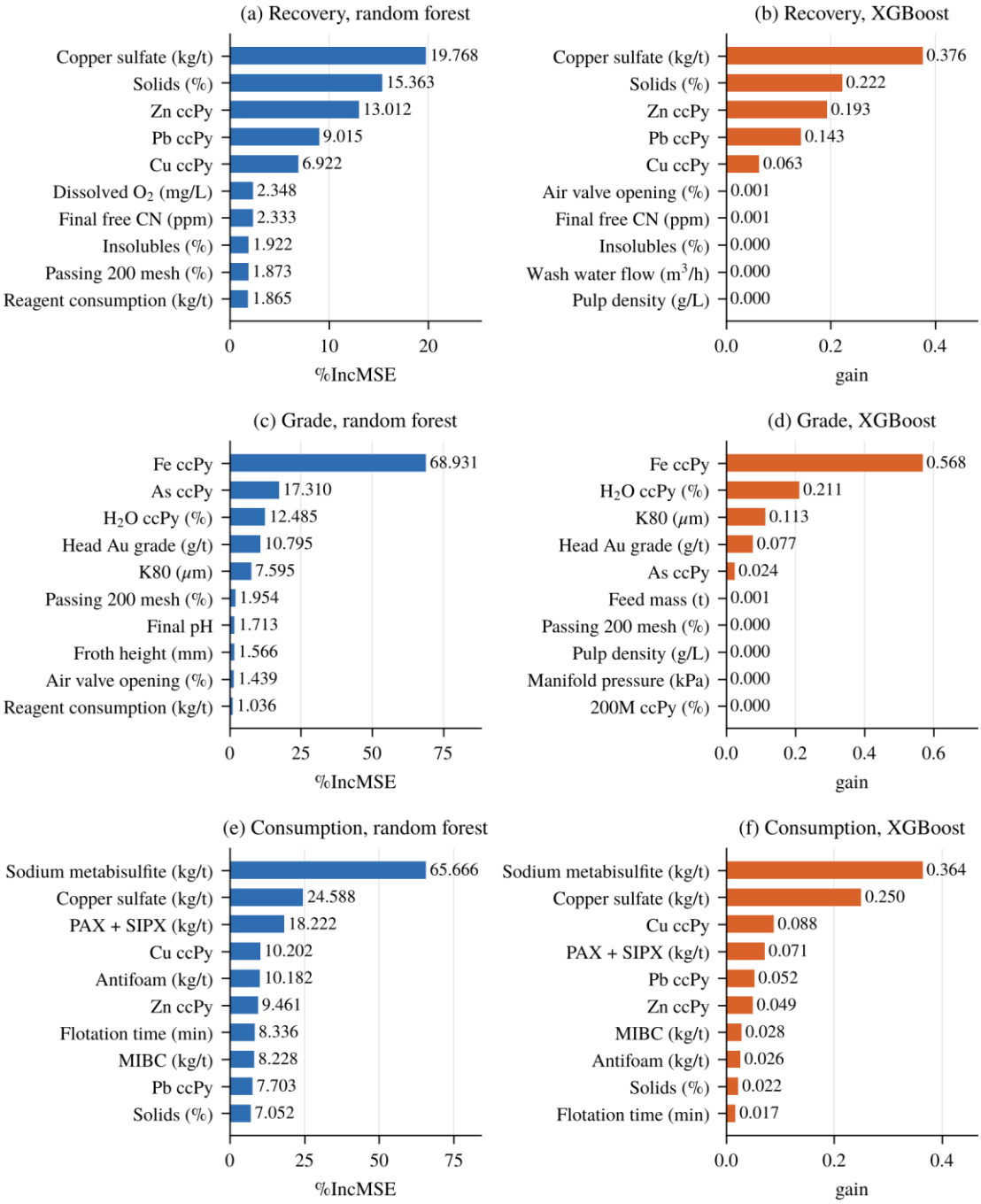


Figure 4: Ten most important predictors for each target and learner: permutation importance (%IncMSE) for the random forest and relative gain for XGBoost. The ccPy labels denote the block of determinations discussed in Section 5.2.

5 Other Issues

This section places the results in the context of prior work and discusses availability, the contrast between the learners, the limits of the statistical tools, and the shift of the process over time.

5.1 Comparison With Prior Flotation Studies

Table 6 summarizes the performance reported by recent flotation studies. The values obtained here lie in the upper band of that range. For recovery they exceed the R^2 of 0.843 of He et al. (2024), the 0.8541 of Erkan and Günen (2026), and the 0.88 of Chehreh Chelgani et al. (2024); only Amankwaa-Kyeremeh et al. (2024) report smaller errors, with more than 1.3 million records against 464 fitting records here. The agreement with Chehreh Chelgani et al. (2024) goes beyond magnitude, since they also found random forests ahead of XGBoost. For grade, our results exceed the R^2 of 0.92 of Devasahayam (2025b) on laboratory data and the 0.54 that Zhao et al. (2026) report for Fe grade, a setting in which random forests and XGBoost were even less accurate. Amadghous et al. (2026) found very small differences among CatBoost, random forests, and XGBoost for Cu grade without testing them, which is consistent with our grade result. For consumption, our errors are larger than those of Yu et al. (2025) and Pérez Cortés et al. (2025), who had 26,225 continuous and 366 daily records, respectively, and did not face a comparable drop in the level of the target between fitting and test.

Study	Data and target	Best model	Reported performance
He et al. (2024)	Processing plant, 182 days; recovery	Stacking (10 variables)	R^2 = 0.843; MAE = 0.929; RMSE = 1.370; MRE = 1.229%
Amankwaa-Kyeremeh et al. (2024)	Cu rougher circuit, 1,325,270 records (6 years); Cu recovery	GPR (Matérn 3/2)	r > 0.96; RMSE < 0.42; MAPE < 0.25%
Erkan and Günen (2026)	Carbonate Zn ores; Zn recovery	Random forest	R^2 = 0.8541
Chehreh Chelgani et al. (2024)	Industrial Cu circuit, 684 daily records; grade and recovery	CatBoost	R^2 up to 0.90 (grade) and 0.88 (recovery); random forest 0.75–0.87; XGBoost 0.64–0.80
Ali et al. (2025)	62 tests on complex sulfides; Pb and Cu recovery and grade	HyFIS	R^2 = 0.9128; RMSE = 2.859
Ji et al. (2024)	641 coal flotation tests; yield and ash	XGBoost	Yield R^2 = 0.836; ash R^2 = 0.853
Devasahayam (2025b)	Laboratory gold flotation; cumulative Au grade	Bayesian ridge (PyMC)	R^2 = 0.92; MSE = 102.37
Nasiri Abarbekouh et al. (2025)	Digital twin of two Au rougher cells; Au grade	PINN	65% lower MSE and 34% lower MRE than the best data-only model
Amadghous et al. (2026)	Polymetallic plant, 620–759 shift samples; Cu, Pb, Zn grades	CatBoost	Cu grade MAE: CatBoost 1.43, random forest 1.46, XGBoost 1.47; not tested
Zhao et al. (2026)	Fe flotation plant, hourly data; Fe and SiO_2 grade	DGraFormer	Fe: R^2 = 0.5423 (persistence 0.4660)
Yu et al. (2025)	Coal plant, 26,225 records; collector and frother dosage	Random forest (single models)	MRE = 5.94% and 8.11%
Pérez Cortés et al. (2025)	Two Cu concentrators, 366 daily records; water consumption	Neural network	MAPE: 4.47% (network); 4.95% (random forest); 5.17% (XGBoost)

Study	Data and target	Best model	Reported performance
This study	Leach-tailings plant, 547 shift records; recovery, grade, consumption	Random forest; XGBoost	R^2 = 0.973, 0.985, 0.897; MAPE = 0.94%, 0.46%, 8.92%

Table 6: Predictive performance reported in recent flotation studies (test or validation data as reported by each study). MRE: mean relative error, comparable to MAPE; PINN: physics-informed neural network; GPR: Gaussian-process regression.

Devasahayam (2025a) found grinding power, head grade, and processing time to dominate gold recovery, whereas here they do not. The difference has an operational explanation: the plant has no grinding stage, fineness is set by the incoming tailings and varied little (88.8% passing 200 mesh in test, standard deviation 3.9 pp), and so did head grade (1.84 g/t, standard deviation 0.24 g/t). As noted in Section 4.1, a predictor that barely changes cannot explain the variation of the target, however real its physical effect.

5.2 Leakage Through Predictor Availability

High accuracy must be read together with the provenance of the predictors. Several features of the records suggest that the ccPy block describes the pyrite concentrate rather than the feed: its average moisture of about 11.7% is typical of a filtered product; its 99.4% passing 200 mesh is finer than the feed (88.8% in test); its Fe grade near 33% is compatible with a pyrite-rich concentrate; and its Ag grade of about 90 g/t is far above that of the Peruvian gold tailings used as references, 29.26 and 31.75 g/t (Berrospi Borda, 2023; Mory Blas, 2024). If this reading is confirmed, the gold grade and the ccPy grades come from the same sample and analytical cycle, and the grade model is a cross-assay calibration in the sense of Section 2.2. It may still be useful—to estimate gold grade from faster or cheaper determinations than fire assay, or to detect inconsistent analyses—but it cannot anticipate the grade, and its metrics overstate what a strict soft sensor would achieve (Kapoor and Narayanan, 2023).

The recovery model is mixed. Copper sulfate and percent solids are recorded during the shift and support an anticipatory component. The Zn, Pb, and Cu grades of the ccPy block, if they describe the concentrate, measure how much base-metal sulfide accompanied the pyrite into the froth, which relates to selectivity and, through the concentrate mass C, to recovery (1); that component becomes available only when the laboratory reports. Figure 4 also shows that the specific consumption, itself one of the targets, was supplied as an input to the recovery and grade models, so its timing deserves the same check. Using either model as a soft sensor in the strict sense, like those of Pural (2023) or Nasiri Abarbekouh et al. (2025), requires documenting the sampling point and reporting time of every predictor and refitting with the variables known before the laboratory result; comparing the two versions would quantify how much of the accuracy depends on concentrate information.

5.3 Why Boosting Wins on an Additive Target

By (2), consumption is the sum of five dosages that are all among the predictors, and the agreement of the subset means in Section 4.5 confirms the identity. The linear predictor with unit coefficients on the five dosages therefore reproduces consumption without error in every period, whatever the level shift. The consumption model, as configured, mainly shows that the procedure recovers a

known relation; to become predictive it would have to estimate the required dosage, or the consumption, from the feed and the operating conditions before dosing, in the spirit of Yu et al. (2025) and Hasidi et al. (2024).

The identity also helps explain why XGBoost beat the random forest so clearly on this target ($p < 0.001$ under both losses). Boosting builds the model as a sum of trees added stagewise (Friedman, 2001), so an additive response can be represented with relative ease, each tree absorbing part of the effect of one reagent. A random forest instead averages trees that each try to approximate the whole function, and its predictions are averages of observed responses:

Lemma 4 *Suppose each tree predicts the average, with bootstrap multiplicities, of the fitting responses in the leaf that contains x. Then $\hat{f}_{\mathrm{RF}}(x) = \sum_i w_i(x)\, y_i$ with $w_i(x) \geq 0$ and $\sum_i w_i(x) = 1$; in particular $\min_i y_i \leq \hat{f}_{\mathrm{RF}}(x) \leq \max_i y_i$.*

Because the weights are spread over the records that share leaves with x, forest predictions contract toward the center of the fitting distribution in regions with few data. That is what happened with the low consumptions of the test period, which were rare in the fitting data: XGBoost overestimated them slightly, and the random forest accumulated an MAE about 60% larger (0.226 against 0.142 kg/t). For recovery the ordering was the opposite. A plausible explanation, which we did not test, is the structure of the noise: shift recovery results from many interactions among feed, pulp chemistry, and operation and carries sampling and assay noise, and averaging weakly correlated trees reduces variance (Breiman, 2001), while stagewise fitting can adapt to part of that noise if regularization does not prevent it.

5.4 Limits of the Diebold-Mariano Test and of Importance Measures

The Diebold-Mariano test assumes a covariance-stationary loss differential, and with 83 records its power is limited; in industrial series the errors may also be autocorrelated or heteroskedastic, conditions under which the standard version can be oversized (Harvey et al., 2024, 2025; Coroneo and Iacone, 2025). The correction of Harvey et al. (1997) mitigates the small-sample problem but does not remove it. The grade result, with $p = 0.0797$ under absolute loss, should be read as insufficient evidence of a difference, not as evidence of equivalence. That the difference was significant under squared but not absolute loss suggests that the random forest's advantage came mainly from fewer large errors, which the squared loss weights more, rather than from a better typical error. The recovery result also carries a methodological lesson: XGBoost was better on validation and the random forest on test, so a selection based only on the validation block would have chosen the learner that later proved inferior. Evaluation schemes that respect time order (Cerqueira et al., 2020) and statistical tests rather than a single ranking (Lones, 2024) guard against such reversals.

The importance measures have their own limits. %IncMSE and gain are global summaries that can be biased by the scale or number of distinct values of a predictor (Strobl et al., 2007) and that, with correlated predictors, can split or concentrate importance in misleading ways (Ewald et al., 2024). The Zn, Pb, and Cu grades of the ccPy block are plausibly correlated, and in XGBoost five predictors accumulated more than 99% of the gain for recovery and grade. Leave-out-covariates parameters (Lei et al., 2018) and their decorrelated versions (Verdinelli and Wasserman, 2024) offer importance targets with inferential guarantees, and SHAP values would decompose individual predictions (Lundberg et al., 2020); both are natural next steps.

5.5 Regime Shift and Concept Drift

The changes in throughput, flotation time, and dosage between periods, together with the fall in consumption and recovery, indicate that the relation between inputs and outputs can move over time, a phenomenon known as concept drift (Lu et al., 2019). A model trained once would lose accuracy as the plant changes feed or practice, so periodic retraining and monitoring of errors and input distributions are conditions for use, not optional complements. Two further limitations concern the protocol. The MAPE threshold, taken from a general scale, was too lenient for targets with large values and low relative variation, and thresholds expressed relative to the baseline, such as ratios of errors or the reductions (4), would be more informative (Hewamalage et al., 2023). And hyperparameters were selected on a single validation block; a rolling-origin scheme would give a more stable estimate (Cerqueira et al., 2020). Finally, 547 records from nine months in one plant support conclusions for that operation but not their transfer to other tailings plants without new validation.

6 Conclusion

On nine months of shift-level data from an industrial plant that reprocesses cyanide-leached gold tailings by flotation, random forests and XGBoost predicted gold recovery, concentrate grade, and specific reagent consumption with test R^2 between 0.897 and 0.985 and error reductions above 85% relative to a historical-mean baseline, under a strictly chronological protocol. No learner dominated: random forests were significantly better for recovery, XGBoost for consumption, and the grade comparison was inconclusive under absolute loss.

The more general lessons concern evaluation. The out-of-sample R^2 of a mean baseline equals minus the squared standardized level shift, so negative baseline values measure how far the process moved, and a learner that clears an R^2 threshold on a later period must track that shift. Absolute MAPE thresholds can be met by a naive predictor and should be replaced by criteria relative to the baseline. Rankings on a single block can reverse between validation and test, which argues for formal pairwise tests under more than one loss. Most importantly, accuracy is only as meaningful as the timing of the predictors: assays made on the same sample as the target and targets that are exact sums of predictors produce excellent metrics that a deployed soft sensor would not reproduce. Refitting with predictors available before the laboratory result, reformulating consumption as a dosage-requirement problem, rolling-origin evaluation, and decorrelated importance measures are the next steps for this plant.

Acknowledgments and Disclosure of Funding

We thank the Graduate School of Universidad César Vallejo and its Master's Program in Artificial Intelligence, our advisor MSc. Ruben Quispe Llacctarimay for his methodological guidance, and the company that operates the concentrator, which is not named for confidentiality reasons, for authorizing the use of its operational and metallurgical records, as well as its operations and laboratory staff for sharing their knowledge of the process. The plant data are confidential and cannot be shared. [Funding sources and competing interests to be declared by the authors.]

Appendix A. Proofs

This appendix proves the results stated in the text. Throughout, $\mathcal{T}$ is the set of m test indices, $\bar{y} = \bar{y}_{\text{test}}$, and $\sigma^2 = \sigma^2_{\text{test}} = m^{-1}\sum_{t\in\mathcal{T}}(y_t - \bar{y})^2$.

Proof of Lemma 1. For any constant c, expanding around $\bar{y}$ and using $\sum_t (y_t - \bar{y}) = 0$,

$$\sum_{t\in\mathcal{T}} (y_t - c)^2 = \sum_{t\in\mathcal{T}} (y_t - \bar{y})^2 + m(\bar{y} - c)^2 = m\sigma^2 + m(\bar{y} - c)^2.$$

Substituting in (3) with $c = \bar{Y}$ gives $R^2 = 1 - \{m\sigma^2 + m(\bar{y} - \bar{Y})^2\}/(m\sigma^2) = -(\bar{y} - \bar{Y})^2/\sigma^2 = -\delta^2$. ■

Proof of Lemma 2. The denominator of (3) equals $m\sigma^2$ and the numerator equals $m\,\text{RMSE}^2$, so $R^2 = 1 - \text{RMSE}^2/\sigma^2$, and solving for RMSE gives the claim. ■

Proof of Lemma 3. By the Cauchy-Schwarz inequality, $\sum_t |e_t| \le \sqrt{m}\,(\sum_t e_t^2)^{1/2}$, which gives $\text{MAE} \le \text{RMSE}$; and $\sum_t e_t^2 \le (\sum_t |e_t|)^2$, which gives $\text{RMSE} \le \sqrt{m}\,\text{MAE}$, with equality when a single error is nonzero. For the normal case, $\text{E}|e| = s\sqrt{2/\pi}$ and $\text{E}(e^2) = s^2$. For the Laplace case, $\text{E}|e| = b$ and $\text{E}(e^2) = 2b^2$. ■

Proof of Lemma 4. Let N_{bi} be the multiplicity of record i in the bootstrap sample of tree b and $\ell_b(x)$ the leaf of tree b that contains x. The tree prediction is

$$T_b(x) = \sum_i v_{bi}(x)\, y_i, \qquad v_{bi}(x) = \frac{N_{bi}\,\mathbf{1}\{x_i \in \ell_b(x)\}}{\sum_k N_{bk}\,\mathbf{1}\{x_k \in \ell_b(x)\}},$$

with $v_{bi}(x) \ge 0$ and $\sum_i v_{bi}(x) = 1$. Averaging over trees gives $w_i(x) = B^{-1}\sum_b v_{bi}(x)$, which inherits both properties, so $\hat{f}_{\text{RF}}(x)$ is a convex combination of the fitting responses and lies between their minimum and maximum. The argument does not apply to boosting, whose predictions are sums of leaf values fitted to successive residuals. ■

Appendix B. Summary of the Predictors

Table 7 gives the mean and standard deviation of the 38 predictors in each chronological subset, grouped by block, together with the number of missing values in the 547 records. The ccPy grades are reported without units, as in the plant records.

Predictor	Training (n = 382)	Validation (n = 82)	Test (n = 83)	NA
Feed and composition				
Cu ccPy	0.889 (0.218)	0.585 (0.261)	0.669 (0.165)	0
Pb ccPy	3.124 (0.678)	2.154 (0.795)	2.471 (0.564)	0
Zn ccPy	4.483 (0.893)	3.138 (1.096)	3.562 (0.610)	0
Fe ccPy	33.007 (4.687)	32.647 (3.525)	32.867 (2.494)	0
As ccPy	6.662 (2.089)	6.581 (1.655)	6.655 (1.151)	0
Ag ccPy	89.720 (12.611)	89.886 (17.241)	89.938 (28.099)	0
H_2O ccPy (%)	11.758 (2.446)	11.614 (1.821)	11.732 (1.239)	0
Passing 200 mesh, ccPy (%)	99.424 (0.285)	99.428 (0.284)	99.438 (0.226)	0
Head Au grade (g/t)	1.850 (0.451)	1.809 (0.347)	1.844 (0.235)	0
Solids (%)	48.379 (3.385)	43.350 (4.140)	44.770 (2.546)	0
Insolubles (%)	14.064 (3.830)	14.394 (3.561)	13.970 (2.681)	0
Pulp density (g/L)	1559.168 (66.646)	1531.344 (55.635)	1582.169 (32.995)	0
Passing 200 mesh, feed (%)	89.041 (6.210)	86.038 (5.184)	88.757 (3.914)	0
K30 (µm)	1.803 (0.053)	1.803 (0.053)	1.803 (0.037)	0
K50 (µm)	10.813 (0.315)	10.827 (0.335)	10.815 (0.260)	0
K80 (µm)	38.476 (8.491)	37.774 (6.262)	38.088 (4.784)	0
Reagent dosage				
Copper sulfate (kg/t)	1.819 (0.405)	1.196 (0.495)	1.368 (0.308)	0
PAX + SIPX (kg/t)	0.360 (0.214)	0.291 (0.103)	0.211 (0.056)	0
MIBC (kg/t)	0.063 (0.040)	0.043 (0.018)	0.046 (0.017)	0
Sodium metabisulfite (kg/t)	0.974 (0.403)	1.101 (0.710)	0.347 (0.431)	0
Antifoam (kg/t)	0.042 (0.108)	0.041 (0.100)	0.042 (0.090)	0
Physicochemical				
Initial pH	9.001 (0.364)	8.825 (0.193)	8.859 (0.171)	0
Final pH	8.612 (0.404)	8.337 (0.235)	8.178 (0.285)	0
Redox potential (mV)	−3.352 (62.757)	13.707 (46.277)	30.982 (48.811)	0
Dissolved O_2 (mg/L)	7.076 (0.564)	7.129 (0.581)	7.074 (0.381)	0
Initial free CN (ppm)	70.527 (39.677)	60.213 (40.925)	54.518 (30.027)	0
Final free CN (ppm)	30.339 (18.933)	15.238 (5.904)	14.307 (4.423)	0
Residual free CN (mg/L)	12.641 (1.695)	12.393 (1.662)	12.602 (1.693)	31
Operational and hydraulic				
Feed mass (t)	471.993 (177.721)	613.224 (150.564)	695.150 (121.886)	0
Pulp flow (m^3/h)	46.463 (17.072)	63.039 (14.527)	66.393 (10.584)	0
Wash-water flow (m^3/h)	0.918 (0.203)	0.925 (0.205)	0.933 (0.144)	0
Pulp level (%)	62.381 (1.865)	62.444 (1.690)	62.414 (1.203)	0
Flotation time (min)	149.681 (81.008)	92.257 (25.577)	75.933 (8.372)	0
Aeration and froth				
Air flow (Nm^3/h)	99.338 (4.238)	99.997 (4.607)	99.461 (3.953)	0
Air pressure (kPa)	35.678 (2.663)	35.976 (2.563)	35.846 (1.979)	0
Manifold pressure (kPa)	50.634 (9.007)	50.768 (10.361)	50.910 (8.151)	0
Air-valve opening (%)	41.671 (2.601)	41.756 (2.925)	41.350 (1.975)	0
Froth height (mm)	384.136 (26.617)	380.488 (27.818)	384.366 (19.701)	0

Table 7: Mean (standard deviation) of the predictors by chronological subset. NA: number of missing values in the 547 records.

References

O. Allard, M. Lopez, I. Demers, and L. Coudert. Gold recovery from sulfide concentrates produced by environmental desulfurization of mine tailings. *Minerals*, 12(8):1011, 2022.

D. Ali, M. B. Hayat, L. Alagha, K. C. Monyake, and H. Khalid. Artificial intelligence and machine learning models for predicting the metallurgical performance of complex sulfide ore flotation process. *Mineral Processing and Extractive Metallurgy*, 134(1):13–32, 2025.

Y. Amadghous, I. Berrada, A. Zagriri, A. Aboulaich, K. Sefrioui Boujemaa, and R. Beniazza. Grade and tailing prediction in polymetallic flotation: DT ensembles vs ANNs on Moroccan industrial data. In *Proceedings of the 12th World Congress on Mechanical, Chemical, and Material Engineering (MCM'26)*, paper MMME 149, London, 2026.

B. Amankwaa-Kyeremeh, C. McCamley, M. Zanin, C. Greet, K. Ehrig, and R. K. Asamoah. Prediction and optimisation of copper recovery in the rougher flotation circuit. *Minerals*, 14(1):36, 2024.

A. Bendaouia et al. Hybrid features extraction for the online mineral grades determination in the flotation froth using deep learning. *Engineering Applications of Artificial Intelligence*, 129:107680, 2024.

C. J. Berrospi Borda. Tratamiento metalúrgico a los relaves para la recuperación de oro a nivel de laboratorio en la planta concentradora Copperfield Perú S.A.C. Lima—2022. Undergraduate thesis, Universidad Nacional Daniel Alcides Carrión, Cerro de Pasco, Peru, 2023.

L. Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

L. Breiman, A. Cutler, A. Liaw, and M. Wiener. *randomForest: Breiman and Cutler's Random Forests for Classification and Regression*, 2024. R package version 4.7-1.2.

C. I. Castellón, N. Toro, E. Gálvez, P. Robles, W. H. Leiva, and R. I. Jeldres. Froth flotation of chalcopyrite/pyrite ore: a critical review. *Materials*, 15(19):6536, 2022.

C. I. Castellón, P. Toledo, F. Rojas, F. D. Sepúlveda, and R. I. Jeldres. Pyrite depression by sodium metabisulfite in freshwater and seawater with copper activation. *Minerals*, 14(10):1036, 2024.

V. Cerqueira, L. Torgo, and I. Mozetič. Evaluating time series forecasting models: an empirical study on performance estimation methods. *Machine Learning*, 109(11):1997–2028, 2020.

W. Chang et al. *shiny: Web Application Framework for R*, 2026. R package version 1.14.0.

S. Chehreh Chelgani, A. Homafar, H. Nasiri, and M. Rezaei Laksar. CatBoost-SHAP for modeling industrial operational flotation variables—A "conscious lab" approach. *Minerals Engineering*, 213:108754, 2024.

T. Chen and C. Guestrin. XGBoost: a scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

T. Chen et al. *xgboost: Extreme Gradient Boosting*, 2026. R package version 3.2.1.1.

X. Chen, D. Liu, L. Yu, P. Shao, M. An, and S. Wen. Recent advances in flotation froth image analysis via deep learning. *Engineering Applications of Artificial Intelligence*, 147:110283, 2025.

F. Concha A. and O. A. Bascur. *The Engineering Science of Mineral Processing: A Fundamental and Practical Approach*. CRC Press, Boca Raton, FL, 2024.

L. Coroneo and F. Iacone. Testing for equal predictive accuracy with strong dependence. *International Journal of Forecasting*, 41(3):1073–1092, 2025.

S. Devasahayam. Interpretable machine learning for identifying key variables influencing gold recovery and grade. *Materials*, 18(18):4318, 2025a.

S. Devasahayam. Advancing flotation process modeling: Bayesian vs. sklearn approaches for gold grade prediction. *Minerals*, 15(6):591, 2025b.

F. X. Diebold and R. S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

I. Erkan and M. A. Günen. Comparison of Zn recovery prediction from carbonate ores with machine-learning methods. *International Journal of Minerals, Metallurgy and Materials*, 33(3):820–832, 2026.

H. Estay, P. Lois-Morales, G. Montes-Atenas, and J. Ruiz del Solar. On the challenges of applying machine learning in mineral processing and extractive metallurgy. *Minerals*, 13(6):788, 2023.

F. K. Ewald, L. Bothmann, M. N. Wright, B. Bischl, G. Casalicchio, and G. König. A guide to feature importance methods for scientific inference. In *Explainable Artificial Intelligence*, Communications in Computer and Information Science, pages 440–464. Springer, Cham, 2024.

J. H. Friedman. Greedy function approximation: a gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232, 2001.

A. Gomez-Flores, S. Ilyas, G. W. Heyes, and H. Kim. A critical review of artificial intelligence in mineral concentration. *Minerals Engineering*, 189:107884, 2022.

L. Grinsztajn, E. Oyallon, and G. Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems 35, Datasets and Benchmarks Track*, 2022.

J. F. Hair, Jr., G. T. M. Hult, C. M. Ringle, and M. Sarstedt. *A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM)*. SAGE, Thousand Oaks, CA, 3rd edition, 2022.

D. Harvey, S. Leybourne, and P. Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, 1997.

D. I. Harvey, S. J. Leybourne, and Y. Zu. Tests for equal forecast accuracy under heteroskedasticity. *Journal of Applied Econometrics*, 39(5):850–869, 2024.

D. I. Harvey, S. J. Leybourne, and Y. Zu. Testing for equal average forecast accuracy in possibly unstable environments. *Journal of Business & Economic Statistics*, 43(3):643–656, 2025.

O. Hasidi et al. Data-driven system for intelligent monitoring and optimization of froth flotation circuits using artificial neural networks and genetic algorithms. *Journal of Process Control*, 103198, 2024.

G. He, M. Liu, H. Zhao, and K. Huang. Ensemble prediction modeling of flotation recovery based on machine learning. *International Journal of Mining Science and Technology*, 34(12):1727–1740, 2024.

H. Hewamalage, K. Ackermann, and C. Bergmeir. Forecast evaluation for data scientists: common pitfalls and best practices. *Data Mining and Knowledge Discovery*, 37(2):788–832, 2023.

T. O. Hodson. Root-mean-square error (RMSE) or mean absolute error (MAE): when to use them or not. *Geoscientific Model Development*, 15(14):5481–5487, 2022.

Z. Huang and T. Yang. Gold recovery from cyanidation tailings using flotation process. *Archives of Metallurgy and Materials*, 67(1):37–43, 2022.

R. Hyndman et al. *forecast: Forecasting Functions for Time Series and Linear Models*, 2026. R package version 9.0.2.

Y. Ji et al. Evaluation of parametric effects of coal flotation based on boosting modeling method. *Physicochemical Problems of Mineral Processing*, 60(6):196385, 2024.

S. Kapoor and A. Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023.

P. Kinnunen, M. Karhu, E. Yli-Rantala, P. Kivikytö-Reponen, and J. Mäkinen. A review of circular economy strategies for mine tailings. *Cleaner Engineering and Technology*, 8:100499, 2022.

S. Kursunoglu. A review on the recovery of critical metals from mine and mineral processing tailings: recent advances. *Journal of Sustainable Metallurgy*, 11:2023–2050, 2025.

C. Lartey, J. Liu, R. K. Asamoah, C. Greet, M. Zanin, and W. Skinner. Effective outlier detection for ensuring data quality in flotation data modelling using machine learning (ML) algorithms. *Minerals*, 14(9):925, 2024.

J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018.

Y. Li et al. Artificial intelligence of mineral processing process: a review of research progress. *Journal of Environmental Chemical Engineering*, 13(5):118313, 2025.

M. A. Lones. Avoiding common machine learning pitfalls. *Patterns*, 5(10):101046, 2024.

J. Lu, A. Liu, F. Dong, F. Gu, J. Gama, and G. Zhang. Learning under concept drift: a review. *IEEE Transactions on Knowledge and Data Engineering*, 31(12):2346–2363, 2019.

S. M. Lundberg et al. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020.

Y. M. Mory Blas. Efecto de colectores secundarios en el tratamiento de relaves para la flotación de pirita aurífera. Undergraduate thesis, Universidad Nacional José Faustino Sánchez Carrión, Huacho, Peru, 2024.

F. Nakhaei, S. Rahimi, and M. Fathi. Prediction of sulfur removal from iron concentrate using column flotation froth features: comparison of k-means clustering, regression, backpropagation neural network, and convolutional neural network. *Minerals*, 12(11):1434, 2022.

M. Nasiri Abarbekouh, S. Iqbal, and S. Särkkä. Physics-informed machine learning for grade prediction in froth flotation. *Minerals Engineering*, 227:109297, 2025.

S. A. Pérez Cortés, E. H. Contreras Moreno, H. Flores Páez, J. P. Hurtado Cruz, and J. A. Jarufe Troncoso. Predictive model for water consumption in a copper mineral concentrator plant located in a desert area using machine learning. *Water*, 17(1):15, 2025.

Y. E. Pural. Developing a data-driven soft sensor to predict silicate impurity in iron ore flotation concentrate. *Physicochemical Problems of Mineral Processing*, 59(5):169823, 2023.

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2026.

scikit-learn developers. DummyRegressor. scikit-learn 1.9.1 documentation, 2026. URL https://scikit-learn.org/stable/modules/generated/sklearn.dummy.DummyRegressor.html.

R. Shwartz-Ziv and A. Armon. Tabular data: deep learning is not all you need. *Information Fusion*, 81:84–90, 2022.

C. Strobl, A.-L. Boulesteix, A. Zeileis, and T. Hothorn. Bias in random forest variable importance measures: illustrations, sources and a solution. *BMC Bioinformatics*, 8:25, 2007.

A. Szmigiel, D. B. Apel, K. Skrzypkowski, L. Wojtecki, and Y. Pu. Advancements in machine learning for optimal performance in flotation processes: a review. *Minerals*, 14(4):331, 2024.

I. Verdinelli and L. Wasserman. Decorrelated variable importance. *Journal of Machine Learning Research*, 25:1–27, 2024.

Y. Yu et al. Study on the applicability of coal flotation reagent dosage prediction models trained based on a large amount of actual industrial on-site data. *International Journal of Coal Preparation and Utilization*, pages 1–15, 2025.

S. Zarbakhsh, A. R. Shahsavar, and M. Soltani. Optimizing PGRs for in vitro shoot proliferation of pomegranate with Bayesian-tuned ensemble stacking regression and NSGA-II: a comparative evaluation of machine learning models. *Plant Methods*, 20(1):82, 2024.

D. Zhang, X. Gao, and H. Wang. Prediction model of iron reverse flotation tailings grade based on multi-feature fusion. *Measurement*, 206:112062, 2023.

L. Zhao, F. Qu, H. Chen, Z. Liu, Y. Ji, and X. Liu. A dynamic relationship-aware approach to flotation concentrate grade prediction using DGraFormer. *Separations*, 13(9):261, 2026.